Heat transfer using active bubble liquid as a coolant

Boris Kichatov,* Vladimir Sudakov,** Alexey Korshunov

Semenov Federal Research Central for Chemical Physics, Moscow, 119991, Russia

ABSTRACT

An active suspension can be used as a heat transfer fluid. An active suspension contains micro-/nanoparticles that can move independently by using energy from the surrounding environment. The mobility of the particles enhances mixing in the fluid, which can contribute to intensified heat transfer. Here, we present the results of an experimental study of transient heat transfer on a horizontal wall using an active bubble fluid as a heat transfer fluid. To create this fluid, TiO_2 nanoparticles are dispersed in an aqueous H_2O_2 solution. When the suspension is irradiated with UV light, photocatalytic reactions cause nanoparticle clusters to become mobile, forming O_2 bubbles in the liquid. The results of the study show that the active bubble fluid has an ambiguous effect on heat transfer intensity. This is because, on the one hand, mobile cluster contribute to intensified heat transfer by mixing the liquid; on the other hand, the reduced thermal conductivity of the coolant - due to the presence of gas bubbles - may actually decrease heat transfer. The pH of the solution has a complex effect on heat transfer, and the nature of this effect depends on the temperature of the liquid. The results obtained in this study may contribute to the development of a new class of coolants based on active fluids.

Keywords: Active fluid, Bubbles, Self-propelled particles, Coolants, Thermal conductivity, Convective heat transfer

Corresponding Authors: * Boris Kichatov - b9682563@yandex.ru; ** Vladimir Sudakov - sudakov6@mail.ru.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

Nomenclature

a	Thermal diffusivity, [m^2/s]	z	Coordinate, [m]
C_o	concentration of H_2O_2 , [-]	*Greek symbols*	
c	Heat capacity, [J/K]	η	Fluid dynamic viscosity, [Pa·s]
D	Cluster size, [m]	λ	Thermal conductivity, [W/(m·K)]
d	Bubble diameter, [m]	ξ	Parameter, [m^2]
l	liquid layer height, [m]	ρ	Density, [kg/m^3]
n	Number concentration of bubbles, [$1/m^3$]	υ	Cluster velocity, [m/s]
q	Heat flux density, [W/m^2]	*Subscripts*	
R	Universal gas constant, [J/(mol·K)]	*ef*	two-phase liquid
Re	Reynolds number, [-]	*g*	gas
T	Absolute temperature, [K]	*l*	liquid
t	Time, [s]	*p*	constant pressure
u_z	Fluid velocity	*w*	wall
z_*	Thickness of temperature layer, [m]	*Abbreviations*	
		HTF	Heat transfer fluid

1. Inroduction

Currently, passive fluids are predominantly used as heat transfer fluids (HTFs). The motion of passive HTF is caused by the supply of energy to its outer boundary. Gases [1-3], liquids [4], two-phase liquids with phase change [5-13], bubbly liquids without phase change [14-21], foams [22-27], suspensions [28, 29] and droplets [30-34] can be used as passive HTFs. Along with passive fluids, active fluids [35] can also be used for cooling systems. Emulsions [36-50] and suspensions [51-57] can be used as active fluids. Particles in the active fluid move independently due to the consumption of energy from the environment. The mechanism of movement of active objects is associated with the action of such factors as the Marangoni flow

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[58], diffusiophoresis [59, 60], thermophoresis [59, 61], electrophoresis [62], chemical magnetic force [63-65], photomagnetic force [66], and quantum vortices [67]. Due to the mobility of active objects, the mixing processes in such liquids can be significantly intensified. Based on this idea, the authors of work [68] proposed using an active suspension as HTF. They showed that the heat transfer coefficient in an active suspension can be increased threefold compared to a passive suspension. The intensity of heat transfer in an active suspension depends on the velocity of active particles and increases with it [69]. Based on theoretical analysis, the authors of work [70] showed that due to the active movement of nanoparticles in a suspension, its effective thermal conductivity can be increased by 10 % compared to the case of using a passive suspension. An experimental study [71] analyzed heat transfer on a wall where an active suspension was used as HTF. The active suspension consisted of mobile MnO_2 microparticles dispersed in an aqueous H_2O_2 solution. During the decomposition of H_2O_2, oxygen bubbles were formed on the surface of the microparticles. The authors of this study found that the heat transfer coefficient in an active suspension can be increased by 76 % compared to the case of using a passive suspension. The study [72] also analyzed heat transfer issues on a wall where an active suspension was used as HTF. When exposed to UV light, photocatalytic reactions occur on the surfaces of nanoparticle clusters. In particular, this study showed that with an increase in the velocity of movement of nanoparticle clusters, the intensity of heat transfer increases. The movement of particles in a suspension can occur not only as a result of chemical reactions, but also under the influence of some external field. If a suspension consists of magnetic particles, these particles can move in the liquid under the influence of a magnetic field. Due to the mobility of magnetic particles, the effective thermal conductivity of the suspension can be significantly increased [73]. By using magnetic fluids, the intensity of heat transfer can be increased by evaporating droplets in the Leidenfrost regime [74-79].

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

In conclusion of this section, we note that active fluids undoubtedly have great potential when used as HTFs [80]. However, it should be noted that they also have certain disadvantages. In particular, if gas bubbles are formed in the active suspension, then, due to the lower thermal conductivity of the gas-liquid mixture compared to a homogeneous liquid, the heat transfer in the active suspension may be even lower than in a passive suspension. Therefore, a more detailed analysis of heat transfer on the wall is necessary when using an active bubble fluid as HTF. Here, we investigate heat transfer on a horizontal wall in a non-stationary setting, using an active suspension as HTF. To create it, an aqueous solution of H_2O_2 with TiO_2 nanoparticles dispersed in it is used. When irradiated with UV light, photocatalytic reactions occur on the surface of the TiO_2 nanoparticle clusters, resulting in the formation of O_2 bubbles. Nanoparticle clusters move in liquid via diffusiophoresis. During the experimental study, the H_2O_2 concentration, pH, liquid temperature, and surfactant concentration were varied, and heat transfer on the adiabatic and heat-releasing walls was examined. The results of the study show that the use of an active bubble fluid can have an ambiguous effect on heat transfer intensity. Simple phenomenological estimates are proposed for a qualitative analysis of the obtained results.

2. Experimental section

Hydrogen peroxide (H_2O_2, 37.0 vol.%) and titanium oxide nanoparticles (TiO_2, nanoparticle size 1-3 nm, rutile, PlasmaChem GmbH) are used to create an active bubble liquid. Acetic acid (≥ 99.7 %, Merck) and sodium hydroxide (≥ 98.0 %, Merck) are used to change the pH of the solution. To create an active bubble liquid, TiO_2 nanoparticles (1 wt.%) were dispersed in an aqueous H_2O_2 solution. The nanoparticles were dispersed using ultrasound. During the experiments, the concentration of the H_2O_2 solution was changed by diluting the initial H_2O_2 solution with deionized water. Acetic acid and sodium hydroxide were used to change the pH of the solution. A PH500T pH meter (MT Measurement, China) was used to measure the pH of

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

the solution. A thermostat (Huber K6-CC, Germany) with an error of $\pm 0.5℃$ was used to maintain the temperature of the solution. To study heat transfer on a horizontal wall, the setup shown in Fig. 1a was used. An electric heater was installed at the bottom of a Petri dish. A ceramic heater (1.5-6 W) with dimensions of $8 \times 12 \times 3\ mm$ was used as a heat-releasing element. A thermocouple (Benetech, GM 1313, K-type, with an error of $\pm 0.5℃$) was installed next to it at a distance of 5 mm. A 5 mm thick H_2O_2 solution was poured into the Petri dish. Temperature measurements were performed on the adiabatic wall. For this purpose, the wall temperature was increased to $33.5℃$, using an electric heater, and then the heater was switched off. Immediately after this, a UV light source (365 nm, 7500 lux) was turned on. In addition, the wall temperature was measured with the constantly working heater. After heating the wall to a temperature $33.5℃$, the UV light source was turned on. The pH of the solution was changed during the experiments. The pH of the initial H_2O_2 solution (37 vol.%) was 4.6. Using acetic acid, the pH of the solution was decreased to 1.9, and with sodium hydroxide, it was increased to 8.1. A thermostat (Huber K6-CC, Germany) with an error of $\pm 0.5℃$, was used to maintain the solution temperature. Each experiment was repeated at least five times. The bubble growth rate was measured using the setup shown in Fig. 1b. A cluster of TiO_2 nanoparticles was placed on a glass plate in H_2O_2. A second quartz plate was placed on top of this liquid layer, at a distance $100\ \mu m$ from the bottom plate. When exposed to UV light, O_2 bubble grew on the cluster surface. To visualize the processes, a microscope (Bresser Biolux SEL) equipped with a video camera (TopCam, 30 fps) was used. Each experiment was repeated at least five times. To study the movement of clusters with bubbles adsorbed on their surfaces, the setup shown in Fig. 1c was used. High-speed filming of this process was performed using a microscope (Bresser Biolux SEL). The average velocity of bubble movement was calculated using the TRACKER program, based on Java.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

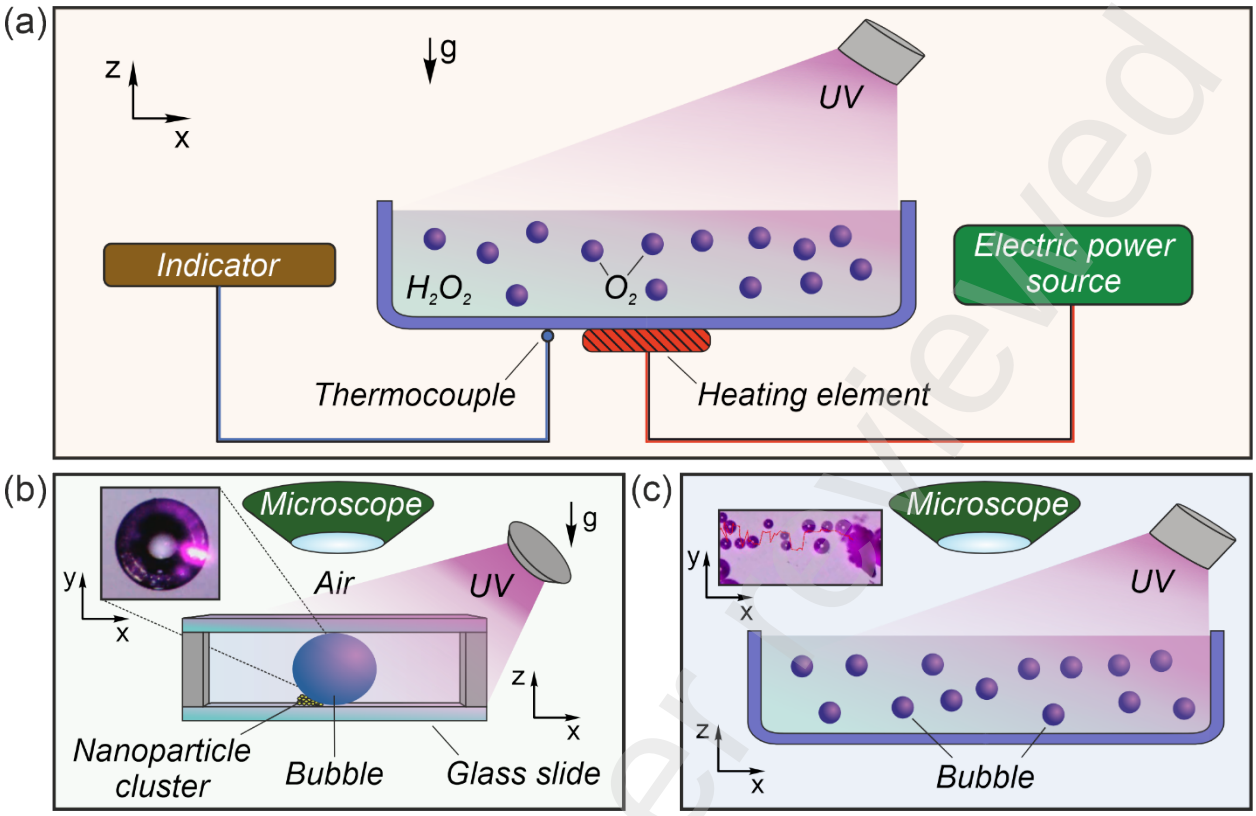


Fig. 1. (a) Schematic diagram of the experimental setup for studying heat transfer. (b) Schematic diagram of the experimental setup for analyzing the bubble growth process. (c) Schematic diagram of the experimental setup for measuring the velocity of nanoparticle clusters.

3. Results and discussion

3.1 Active bubble liquid

In this study, an active bubble fluid is used as HTF. To create it, TiO_2 nanoparticles were dispersed in an aqueous H_2O_2 solution. As a result of nanoparticle agglomeration, microscopic clusters were formed. The attraction between nanoparticles is due to hydrophobic and van der Waals interactions. When irradiated with UV light, photocatalytic reactions $2H_2O_2=2H_2O+O_2$ occur on the surface of the clusters, leading to the formation of O_2 bubbles (Fig. 2c, 2g, Video S1). Clusters with bubbles adsorbed on their surfaces move in the H_2O_2 solution (Video S2). Let us consider the nature of the mechanism responsible for the movement of nanoparticle clusters.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

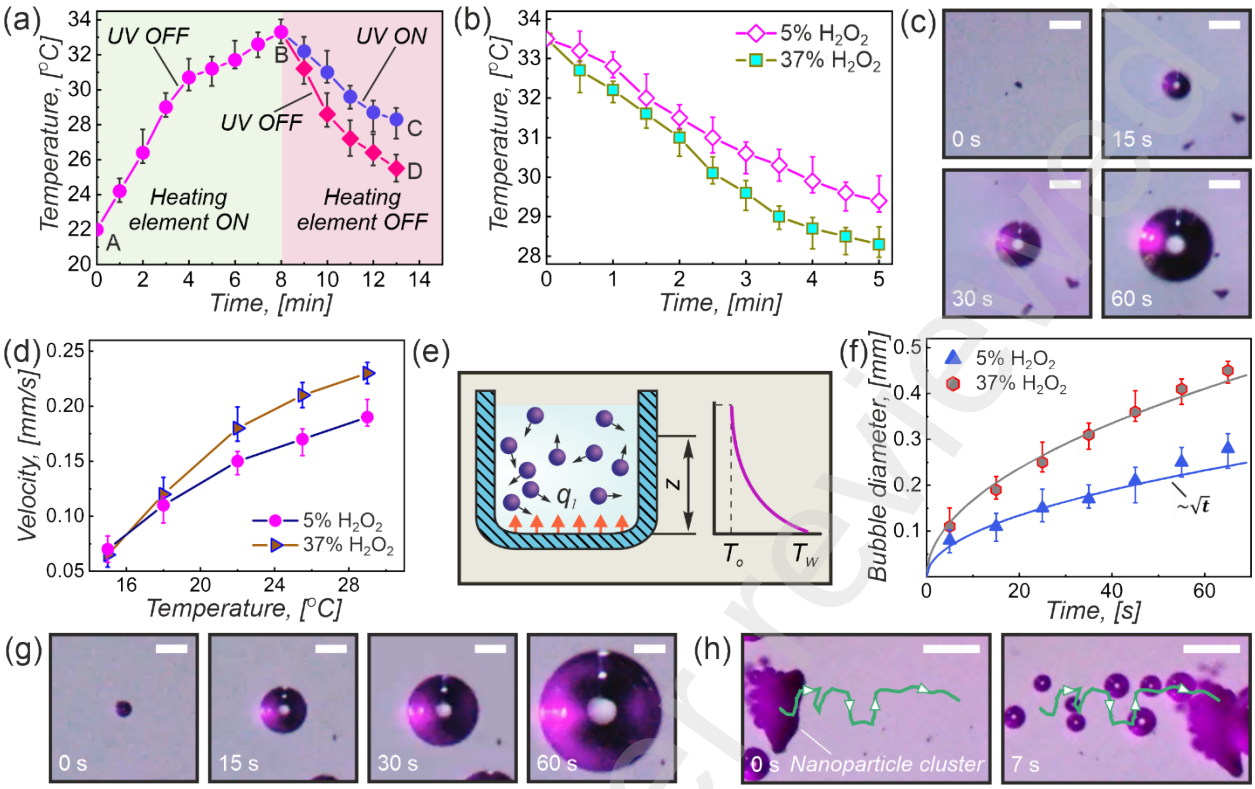


Fig. 2. (a) Time dependence of wall temperature (H_2O_2 concentration 14 vol. %). (b) Effect of H_2O_2 concentration on adiabatic wall temperature. (c), (g) Sequence of images demonstrating bubble growth process: (c) H_2O_2 concentration 5 vol. %; (g) 37 vol. %. Scale bars - 200 μm. (d) Effect of liquid temperature and H_2O_2 concentration on cluster velocity. (e) Schematic diagram of adiabatic wall cooling. (f) Effect of H_2O_2 concentration on bubble growth rate (liquid temperature 22 °C). (h) Sequence of images demonstrating active cluster motion. Scale bars - 500 μm.

During photocatalytic reactions, a concentration boundary layer is formed within the liquid. Across the thickness of this layer the concentration of H_2O_2 changes from its value in the bulk of the liquid to a certain value on the cluster surface. If the cluster is stationary, the thickness of this layer does not change around the cluster. However, if the cluster is moving, the distribution of the thickness of the concentration layer around it becomes asymmetric [56]. In the frontal region of a moving cluster, the thickness of the concentration boundary layer is smaller than in the aft region. As a result, the concentration of H_2O_2 in the frontal region of the cluster will be greater than in the aft region. Due to the fact that a local gradient of H_2O_2 concentration arises near the surface of the cluster, it will move under the action of diffusiophoresis [56]. In this

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

case, the speed of movement of the cluster can be represented as $\upsilon \sim RT\xi C_o \eta^{-1} D^{-1}$ [56]. From this formula it follows that with an increase in the concentration of H_2O_2 and the temperature of the liquid, the speed of movement of the cluster increases, which is consistent with the experimental data (Fig. 2d). It should be noted that despite the fact that gas bubbles are formed on the surface of the cluster, the contribution of the reactive force to its movement is insignificant compared to diffusiophoresis [56].

3.2. Heat exchange on the wall

To study heat transfer on a horizontal adiabatic wall, an experimental setup schematically shown in Fig. 1a was used. A heat-releasing element and a thermocouple were mounted at the bottom of a Petri dish. The dish was filled with H_2O_2 solution containing dispersed TiO_2 nanoparticles. The heater was then turned on, and the wall temperature increased to 33.5℃ (line AB) (Fig. 2a). Immediately after this, the heater was turned off, but a UV light source was turned on. During the photocatalytic reactions, the clusters became mobile (Fig. 2h). The experimental results show that when using an active bubbly liquid as a heat transfer medium, the wall temperature (line BC) is higher compared to the case of using a passive suspension (line BD) (Fig. 2a). Let us consider the cause of this phenomenon. Mobile clusters mix the liquid and should thereby lead to intensification of heat transfer. However, due to the formation of gas bubbles, heat transfer in the liquid by conduction decreases. The competition between these processes means that active HTF can have an ambiguous effect on heat transfer intensity. Let us examine the heat transfer mechanism in more detail. To estimate the wall temperature, we will use the heat balance equation

$$\rho_w h c_w \frac{dT_w}{dt} = q_w - q_l \quad (1)$$

Let us estimate the heat flux transferred from the wall to the fluid. We assume that convection and conduction are the primary mechanisms of heat transfer. It is also assummed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

that the main contribution to the convective motion of the fluid arises from moving bubbles. If a unit volume of fluid contains n bubbles, and their average diameter is equal to d, and they move at a velocity v, then this will lead to a fluid flow with a velocity $u_z \sim nd^3v$. We will assume that the fluid flow is laminar, and the Reynolds number satisfies the criterion $Re \ll 1$. Estimating the orders of the terms in the heat balance equation for a gas-liquid mixture, we obtain

$$u_z \frac{\delta T}{z_*} \sim a_{ef} \frac{\delta T}{z_*^2} \tag{2}$$

From Eq. (2), taking into account the expression for the velocity of convective flow of liquid, one can obtain an estimate of the thickness of the temperature boundary layer

$$z_* \sim \frac{a_{ef}}{u_z} \sim \frac{a_{ef}}{nd^3v} \tag{3}$$

The heat flux that is transferred from the wall to the liquid is equal to

$$q_l \sim \lambda_{ef} \frac{\delta T}{z_*} \sim \frac{a_{ef} \delta T}{\rho_{ef} c_{ef} z_*} \tag{4}$$

As a result of substituting Eq. (3) into Eq. (4), we obtain

$$q_l \sim \rho_{ef} nd^3 v c_{ef} (T_w - T_o) \tag{5}$$

Let us consider heat transfer on an adiabatic wall $q_w = 0$. Substituting Eq. (5) into Eq. (1), we obtain

$$\rho_w h c_w \frac{dT_w}{dt} \sim -\rho_{ef} nd^3 v c_{ef} (T_w - T_o) \tag{6}$$

By integrating Eq. (6), taking into account the initial condition $t = 0$: $T_w = T_w(0)$, we obtain an estimate for the wall temperature.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

$$T_w \sim -[T_w(0) - T_o] exp(-\alpha t) + T_o$$

(7)

$$\alpha \equiv nd^3 \upsilon \rho_{ef} c_{ef} (\rho_w h c_w)^{-1}$$

Eq. (7) describes the change in temperature of the adiabatic wall with time. Let us analyze the effect of the volumetric content of gas in the liquid φ on the wall temperature. Considering that $\varphi \sim nd^3$, and also taking into account the expressions for the effective density of the two-phase mixture $\rho_{ef} = \rho_l(1-\varphi) + \rho_g \varphi$ and the heat capacity $c_{ef} = c_l(1-\varphi) + c_g \varphi$ and assuming that the inequalities $\rho_l \gg \rho_g$ and $c_l \gg c_g$ are satisfied, the following conclusion can be drawn from Eq. (7). The volumetric content of gas in an active bubbly liquid has an ambiguous effect on the temperature of the adiabatic wall. If the volumetric content of gas is less than a certain critical value $\varphi < \varphi_* \approx 0.33$, then an increase in the volumetric content of gas in the liquid will contribute to a decrease in the wall temperature. The reason for this phenomenon is due to the fact that with an increase in the numerical concentration of mobile bubbles in the liquid, mixing processes intensify. At the same time, if the volumetric content of gas in the liquid exceeds a certain critical value $\varphi > \varphi_* \approx 0.33$, then the intensity of heat transfer will decrease along with an increase in the volumetric gas content in the liquid due to a decrease in the thermal conductivity of the gas-liquid mixture. In the case shown in Fig. 2a, the volumetric content of gas in the liquid was approximately $\varphi \approx 0.55$. Therefore, when using an active bubble liquid, the heat transfer intensity was lower than when using a passive suspension.

3.3. Effect of H_2O_2 concentration

From Eq. (7) it follows that the velocity of movement of clusters with bubbles adsorbed on their surface affects the temperature of the adiabatic wall. Considering that the velocity of movement of clusters depends on the concentration of H_2O_2 (Fig. 2d) and taking into account formula for velocity, from Eq. (7) it is possible to obtain an estimate $T_w \sim exp(-const C_o)$. It

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

follows from this that by changing the concentration of H_2O_2 in the solution, it is possible to change the temperature of the adiabatic wall. The experimental results show that with an increase in the concentration of H_2O_2 in the solution, the temperature of the adiabatic wall decreases (Fig. 2b). The reason for this phenomenon is due to the increase in the velocity of movement of bubbles (Fig. 2d) and their diameter (Fig. 2f), which ultimately affects the intensity of convective heat transfer.

3.4. Convective heat transfer

Let us consider the case where convection makes the main contribution to heat transfer, and heat transfer by conduction can be neglected. This case is typical for low values of volumetric gas content in the liquid and high cluster velocities. Given this assumption, the expression for the heat flux transferred from the wall to the liquid can be represented as [72]

$$q_l \sim \upsilon d^3 n \rho_l c_p z \frac{dT}{dz} \tag{8}$$

The right-hand side of Eq. (8) represents the difference in enthalpy fluxes at the wall and in a certain section z (Fig. 2e). Mobile bubbles cannot approach the wall at a distance closer than d. Therefore, integrating Eq. (8) over the range from d to l, we obtain

$$q_l \sim \upsilon d^3 n \rho_l c_p \left(ln \frac{l}{d}\right)^{-1} (T_w - T_o) \tag{9}$$

As a result of substituting Eq. (9) into Eq. (1), we can obtain the following equation

$$\rho_w h c_w \frac{dT_w}{dt} \sim q_w - \rho_l n d^3 \upsilon c_p (T_w - T_o) \tag{10}$$

Let us consider heat exchange on an adiabatic wall $q_w = 0$. Let the wall temperature at the initial moment of time $t = 0$ be equal to $T_w = T_w(0)$. As a result of integrating Eq. (9), we obtain

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

$$T_w \sim -[T_w(0) - T_o] exp(-\beta t) + T_o$$

(11)

$$\beta \equiv nd^3 \upsilon \rho_l c_p [\rho_w h c_w ln(l/d)]^{-1}$$

Given that $\varphi \sim nd^3$, Eq. (11) yields an estimate for the adiabatic wall temperature $T_w \sim exp(-const\varphi)$. This estimate implies that, as the volumetric gas content of the liquid increases, the wall temperature decreases monotonically. This case differs fundamentally from the previously considered case, where the volumetric gas content of the liquid had an ambiguous effect on the heat transfer rate. This peculiarity stems from the fact that, in the case under consideration, heat transfer by conduction does not play a significant role.

3.5. Effect of pH and temperature of the solution

Experimental results show that the adiabatic wall temperature depends on the pH of the solution, and this effect is ambiguous (Fig. 3a-3c). As the pH of the solution decreases, the wall temperature decreases if the initial temperature of the liquid exceeds > 22 ℃ (Fig. 3b, 3c). However, if this condition is not met, then with an increase in the pH of the solution, the wall temperature decreases (Fig. 3a). The ambiguous effect of the pH of the solution on the temperature of the adiabatic wall is due to the ambiguous effect of the pH of the solution on the velocity of gas bubbles (Fig. 3d, 3e, 3f). In addition, if the liquid temperature is > 22 ℃, and pH = 8.1, then the decomposition reaction of H_2O_2 is exothermic, which contributes to an increase in the wall temperature at the initial moment of time (Fig. 3b, 3c). Note that the pH of the solution has a relatively weak effect on the growth rate of O_2 bubbles (Fig. 3g, 3i, 3h, 3j). During the experimental study, the initial liquid temperature was also varied. The experimental results show that there is an optimal liquid temperature 22 ℃ at which the intensity of heat transfer is greatest (Fig. 4e). As the liquid temperature deviates from this value in either direction, the intensity of heat transfer decreases. This phenomenon is caused by the fact that at

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

low liquid temperatures, the growth rate of O_2 bubbles (Figs. 4a, 4b, 4c, 4d) and their velocity of movement (Fig. 3d) are low, which leads to a decrease in the intensity of liquid mixing.

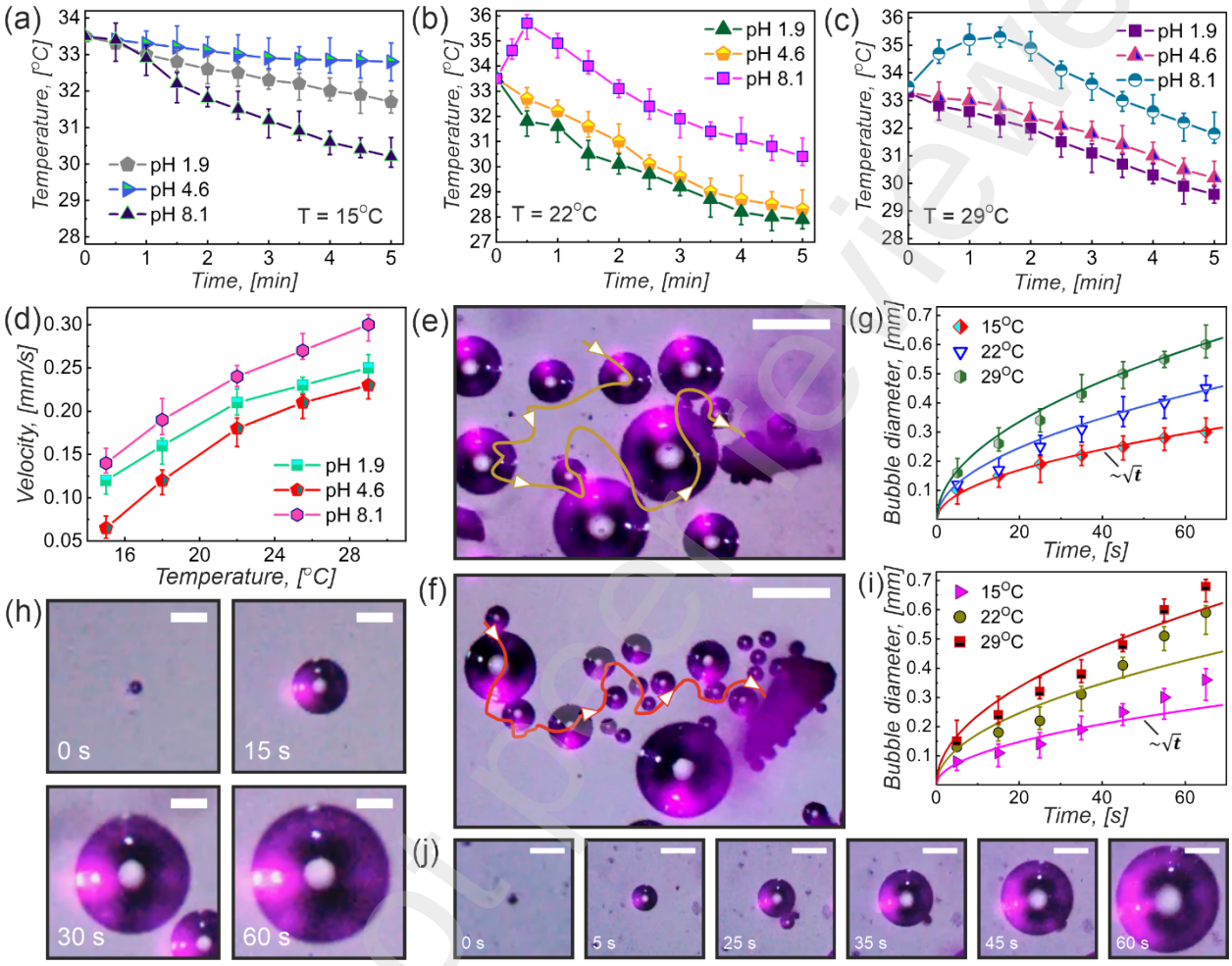


Fig. 3. (a), (b), (c) Effect of solution pH on the wall temperature for different values of the initial liquid temperature: (a) liquid temperature 15 ℃; (b) 22 ℃; (c) 27 ℃. (d) Effect of liquid temperature and solution pH on the velocity of cluster movement. (e), (f) Movement of an active cluster in liquid (37 vol. % H_2O_2, 22 ℃): (e) pH=1.9; (f) pH=8.1. Scale bars - 200 μm. (g), (i) Change in bubble diameter with time for different values of liquid temperature: (g) 37 vol. % H_2O_2, pH=1.9; (i) 37 vol. % H_2O_2, pH=8.1. (h), (j) Sequence of images demonstrating the process of O_2 bubble growth (37 vol. % H_2O_2, 22 ℃): (h) pH=1.9; (j) pH=8.1. Scale bars - 100 μm.

At the same time, at higher liquid temperatures, the temperature difference between the wall and the liquid decreases, which also negatively affects heat transfer. Due to the competition

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

between these processes, there is a certain optimal liquid temperature at which the intensity of heat transfer is greatest.

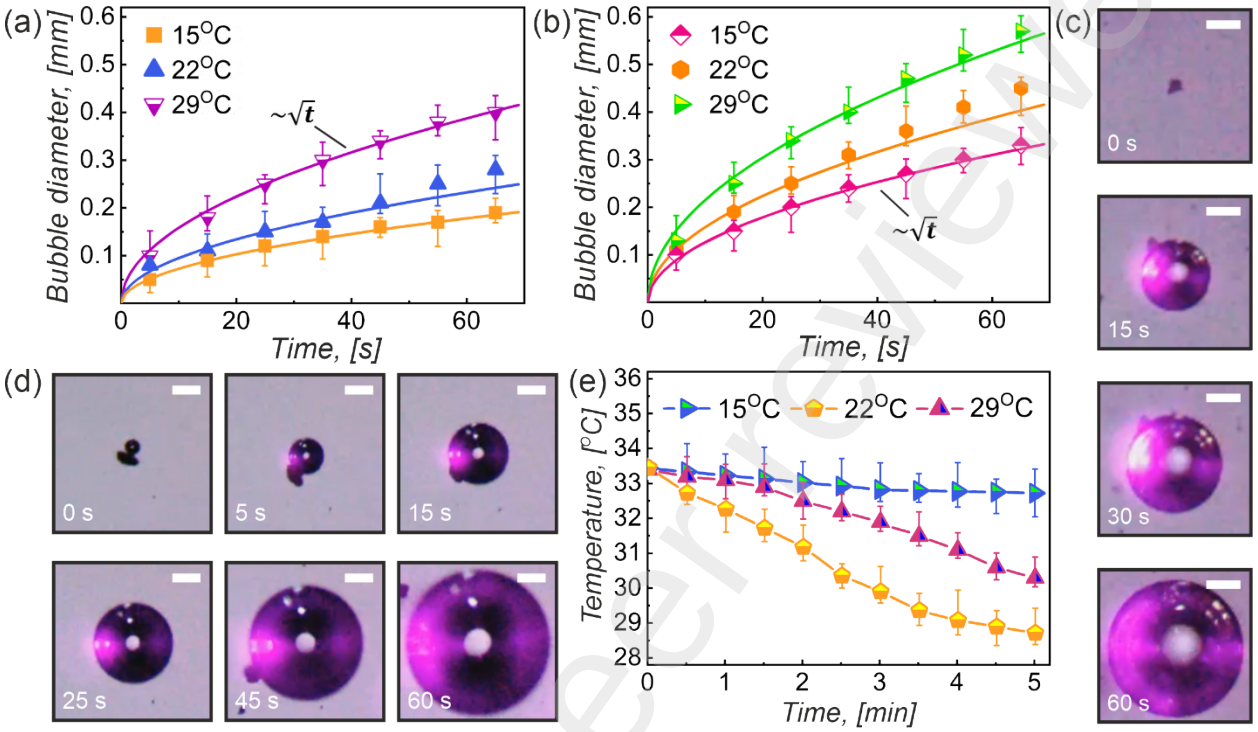


Fig. 4. (a), (b) Change in bubble diameter over time for different values of liquid temperature: (a) 5 vol. % H_2O_2; (b) 37 vol. % H_2O_2. (c), (d) Sequence of images showing the growth of O_2 bubbles with varying H_2O_2 concentrations (29 ℃): (c) 5 vol. % H_2O_2; (d) 37 vol. % H_2O_2. Scale bars - 100 μm. (e) Effect of initial liquid temperature on wall temperature (37 vol. % H_2O_2).

3.6. Surfactant impact

In the absence of surfactant in the solution, bubbles detached from the cluster surface rise under the action of the Archimedes force and collapse at the liquid-air interface. In this case, a certain quasi-steady-state value of the volumetric gas content is maintained within the gas-liquid mixture. However, this picture changes dramatically if a surfactant is present in the solution. In this case, bubbles do not collapse at the liquid-air interface, resulting in the formation of a stable foam (Fig. 5c). From Fig. 5a it can be seen that the presence of surfactant in the solution increases the temperature of the adiabatic wall. Moreover, with increasing SDS

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

concentration in the solution, the wall temperature increases. This is due to the low thermal conductivity of the foam, the low mobility of bubbles in the foam, and small diameter of bubbles (Figs. 5f, 5g, 5h).

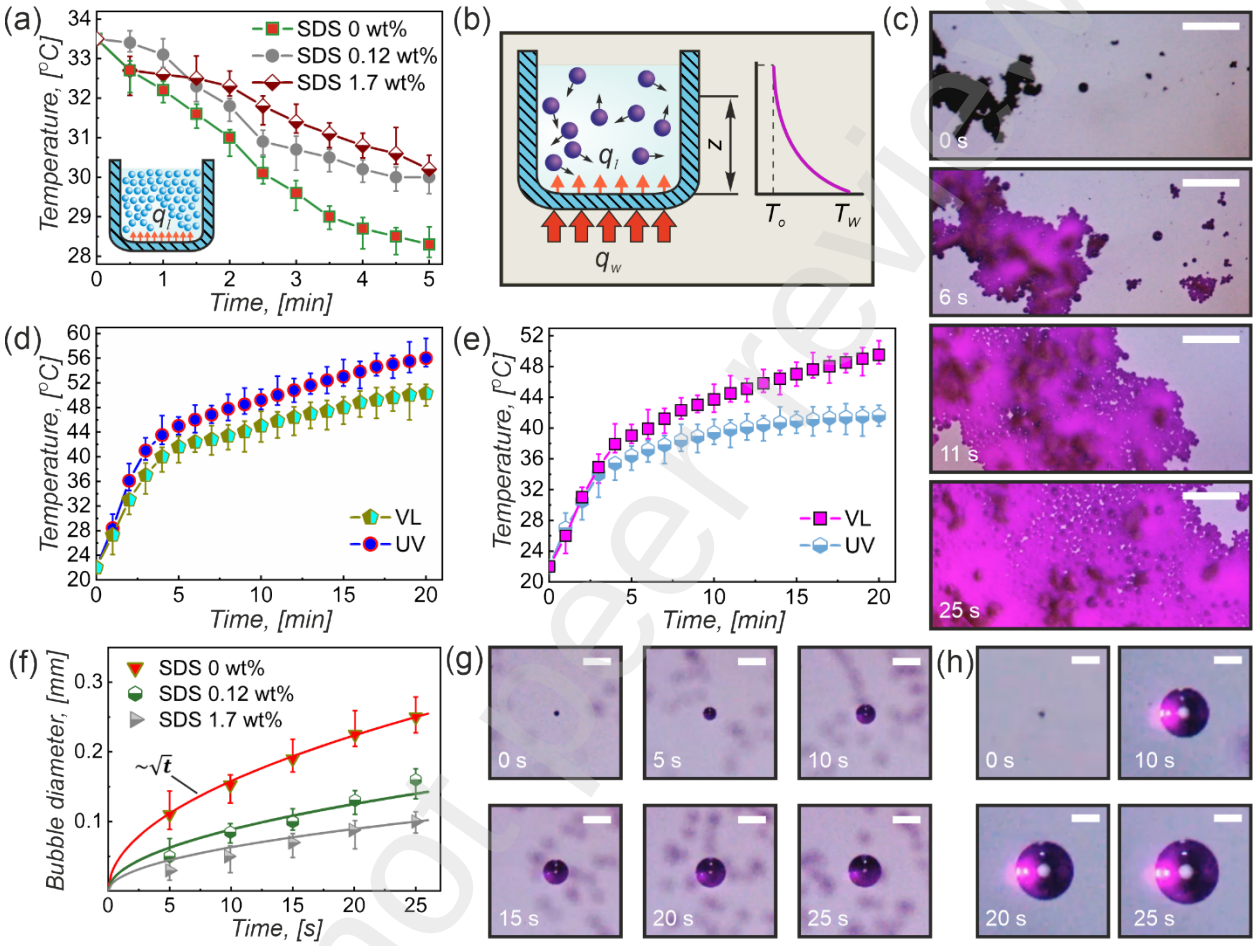


Fig. 5. (a) Effect of SDS on wall temperature (37 vol. % H_2O_2, initial liquid temperature 22 ℃). (b) Heat transfer on the heat-releasing surface. (c) Sequence of images demonstrating the process of formation of a stable foam layer in the active bubbly liquid (37 vol. % H_2O_2, 22 ℃, 1.7 wt. % SDS). Scale bars - 500 *μm*. (d), (e) Change in wall temperature over time: (d) 5 vol. % H_2O_2; (e) 37 vol. % H_2O_2. (f) Effect of SDS concentration on the growth rate of gas bubbles (37 vol. % H_2O_2, 22 ℃). (g), (h) Sequence of images demonstrating the process of O_2 bubble growth with varying SDS concentration (37 vol. % H_2O_2): (g) 1.7 wt. % SDS, (h) 0 wt. % SDS. Scale bars - 100 *μm*.

3.7. Heat exchange on a heat-releasing surface

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

Let us consider the case of heat exchange on a heat-releasing surface $q_w > 0$ (Fig. 5b). When the wall temperature reached 32.5 ℃, a UV light source was turned on. The experimental results show that the use of active bubbly fluid can both increase heat transfer intensity (Fig. 5e) and decrease it (Fig. 5d). The ambiguous effect of active bubbly fluid on wall temperature is due to the competition between convective heat transfer and thermal conductivity. Let us estimate the wall temperature in the case where $q_w > 0$. Integrating Eq. (1), taking into account Eq. (5), and the initial condition $t = 0$: $T_w = T_w(0)$, we obtain

$$T_w \sim [T_w(0) - T_o - \gamma] exp(-\alpha t) + T_o + \gamma \quad (12)$$

$$\alpha \equiv \varphi v \rho_{ef} c_{ef} (\rho_w h c_w)^{-1}$$

$$\gamma \equiv q_w (\varphi v \rho_{ef} c_{ef})^{-1}$$

From Eq. (12), it follows that mobile bubble clusters can have an ambiguous effect on wall temperature. On the one hand, mobile clusters promote liquid mixing and thereby intensify heat transfer. However, on the other hand, the presence of gas bubbles in the liquid reduces heat transfer via thermal conductivity. The competition between these processes leads to the ambiguous effect of active bubbles on wall temperature. The highest heat transfer intensity was observed in the case of using 37 vol. % H_2O_2 solution (Fig. 5e), which is due to the higher velocity of cluster movement in the liquid compared to the case when a 5 vol. % H_2O_2 solution was used (Fig. 5d).

4. Conclusion

Active suspensions can be used as HTFs. Active suspensions contain particles that can move independently by absorbing energy from the surrounding environment. Due to the mobility of active particles, heat transfer intensity can be increased compared to a passive suspension. However, if gas bubbles form during particle movement, heat transfer in such fluid may

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

actually decrease compared to a passive suspension. Based on the data obtained, the following conclusions can be drawn.

(1) An active bubbly liquid can have an ambiguous effect on heat transfer intensity. If the volumetric gas content in the liquid is less than 33 vol.%, mobile clusters will promote heat transfer intensification. However, if this condition is not met, heat transfer in an active bubbly liquid will be lower than in a passive suspension.

(2) The ambiguous effect of an active bubbly liquid on heat transfer intensity is due to the competition between convective heat transfer and conduction. On the one hand, mobile nanoparticle clusters mix different layers of the liquid and thereby promote heat transfer intensification. However, on the other hand, O_2 bubbles reduce the intensity of heat transfer through conduction.

(3) The pH of the solution has an ambiguous effect on heat transfer. This is due to the ambiguous effect of the pH of the solution on the velocity of cluster movement in the liquid. When the liquid temperature is 15 ℃, the highest heat transfer intensity can be achieved at pH = 8.1. However, as the liquid temperature increases > 22 ℃, the highest heat transfer intensity corresponds to pH =1.9.

(4) If the active HTF contains a surfactant, this leads to a decrease in the heat transfer intensity compared to cases in which it is absent. The negative effect of surfactant on heat transfer is due to the formation of quasi-stable foam.

(5) The concentration of H_2O_2 in the solution has an ambiguous effect on the adiabatic wall temperature. This is due to the fact that with increasing H_2O_2 concentration in the solution, the velocity of nanoparticle clusters increases, which entails a decrease in the wall temperature $T_w \sim exp(-constC_o)$. However, if the volumetric gas content in the two-phase mixture exceeds the

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

critical value (33 vol. %), then with increasing H_2O_2 concentration in the solution, the heat transfer intensity will decrease due to a decrease in heat transfer by conduction.

CRediT authorship contribution statement

Boris Kichatov: Conceptualization, Investigation, Project administration, Writing - original draft. **Alexey Korshunov:** Conceptualization, Methodology, Writing – original draft. **Vladimir Sudakov:** Conceptualization, Formal analysis, Investigation, Validation, Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was carried out as part of the state assignment on the topic “Interdisciplinary approaches to the creation and study of micro-/nanostructured systems” (registration no. 125012200595-8).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:

Data availability

Data will be made available on request.

References

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[1] A. Leontiev, A. Saveliev, B. Kichatov, A. Kiverin, A. Korshunov, V. Sudakov, Effect of gaseous coolant temperature on the transpiration cooling for porous wall in the supersonic flow, Int. J. Heat Mass Transfer 142 (2019) 118433. https://doi.org/10.1016/j.ijheatmasstransfer.2019.118433.

[2] W. Chen, Y. Wang, S. Tu, T. Chu, H. Su, K. Wang, Numerical investigation of flow dynamics and cooling performance in combined porosity structures at the leading edge of a hypersonic vehicle, Int. J. Heat Mass Transfer 245 (2025) 126996. https://doi.org/10.1016/j.ijheatmasstransfer.2025.126996.

[3] A. Leontiev, B. Kichatov, A. Korshunov, V. Gubernov, A. Kiverin, Effect of the flow rate of injected air on the temperature of porous wall in the supersonic flow, Int. J. Heat Mass Transfer 175 (2021) 121382. https://doi.org/10.1016/j.ijheatmasstransfer.2021.121382.

[4] P. Bhandari, K.S. Rawat, Y.K. Prajapati, D. Padalia, L. Ranakoti, T. Singh, Design modifications in micro pin fin configuration of microchannel heat sink for single phase liquid flow: A review, J. Energy Storage 66 (2023) 107548. https://doi.org/10.1016/j.est.2023.107548.

[5] S. Ahmadi, M. Eraghubi, M.A. Akhavan-Behabadi, P. Hanafizadeh, S. Sayadian, A.J. Robinson, EHD augmented heat transfer during upward bubbly flow boiling for low to medium frequency AC electric fields, Int. J. Therm. Sci. 191 (2023) 108346. https://doi.org/10.1016/j.ijthermalsci.2023.108346.

[6] A.H. Al-Zaidi, M.M. Mahmoud, A. Ivanov, T.G. Karayiannis, Bubble nucleation site density, generation frequency and departure diameter in flow boiling of HFE-7100, Int. J. Heat Mass Transfer 242 (2025) 126830. https://doi.org/10.1016/j.ijheatmasstransfer.2025.126830.

[7] B. Chen, Y. Ding, M. Zhang, X. Zhang, An empirical study on condensation process and heat transfer enhancement under corona discharge: A visualization-based investigation, Int. Commun. Heat Mass Transfer 158 (2024) 107919.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

https://doi.org/10.1016/j.icheatmasstransfer.2024.107919.

[8] M.B. Darshan, M. Magnini, O.K. Matar, Numerical simulations of flow boiling in constricted microchannels: Bubble dynamics and heat transfer characteristics, Int. J. Heat Mass Transfer 253 (2025) 127456. https://doi.org/10.1016/j.ijheatmasstransfer.2025.127456.

[9] M. Edalatpour, N.T. Agathocleous, J. Byun, J.G. Tasiopoulos, C.W. Engberg, W. Abate, Two-phase computational modeling of evaporative film boiling in plate heat exchangers with offset-strip fins, Int. J. Heat Mass Transfer 236 (2025) 126260.
https://doi.org/10.1016/j.ijheatmasstransfer.2024.126260.

[10] B. Li, H. Yan, Y. He, Z. Zhang, W. Ling, M. Zeng, Y. Wang, Q. Wang, Investigation of the insoluble impurities' impact on flow boiling heat transfer in aluminum microchannel heat exchangers, Energy 311 (2024) 133312. https://doi.org/10.1016/j.energy.2024.133312.

[11] I. El Mellas, N. Samkhaniani, C. Falsetti, A. Stroh, M. Icardi, M. Magnini, Numerical investigation of bubble dynamics and flow boiling heat transfer in cylindrical micro-pin-fin heat exchangers, Int. J. Heat Mass Transfer 228 (2024) 125620.
https://doi.org/10.1016/j.ijheatmasstransfer.2024.125620.

[12] S. Ni, Y. Li, D. Ji, X. Zhang, J. Zhao, Effect of surfactants on flow boiling heat transfer and bubble dynamic: an experimental study, Int. J. Heat Mass Transfer 255 (2026) 127851.
https://doi.org/10.1016/j.ijheatmasstransfer.2025.127851.

[13] F.J. Santos, L. Beckedorff, T.S. Possamai, K.V. Paiva, J. L.G. Oliveira, Two-phase flows downstream, upstream and within Plate Heat Exchangers, Int. J. Multiphase Flow 182 (2025) 105062. https://doi.org/10.1016/j.ijmultiphaseflow.2024.105062.

[14] E.H. Abdel-Gawad, I.H. Saleh, G.H. Sedahmed, M.H. Abdel-Aziz, E.-S. Z. El-Ashtoukhy, A.S. Fathalla, M.A. El-Naggar, Mass and heat transfer behavior of a bubble column fitted with horizontal spiral tubes in bubbly-flow regime, Int. Commun. Heat Mass Transfer 151 (2024) 107258. https://doi.org/10.1016/j.icheatmasstransfer.2024.107258.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[15] M. Guesmi, J. Manthey, R. Schab, S. Unz, M. Beckmann, Mechanistic modelling of water-oxygen bubbly flow in horizontal pipes: Deviation analysis from experimental correlations and performance comparison with CFD, Int. J. Thermofluids 25 (2025) 101016. https://doi.org/10.1016/j.ijft.2024.101016.

[16] L.T. Siow, J.R. Lee, A. Kitagawa, E.H. Ooi, E.V. Lau, Convection heat transfer enhancement of a vertically heated surface via rising bubbles in graphene oxide (GO) nanofluid, Int. J. Heat Mass Transfer 252 (2025) 127450. https://doi.org/10.1016/j.ijheatmasstransfer.2025.127450.

[17] R. Chen, C. Cheng, Z. Liu, Y. Hu, J. Zhao, Z. Zuo, J. Xu, H. Wang, Heat transfer characteristics during bottom-blown bubble coalescence, Int. J. Heat Mass Transfer 233 (2024) 126005. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126005.

[18] G. Huang, J. Han, L. Shan, W. Li, The impacts of gas-liquid two-phase flow regimes on heat transfer performance and pressure drop in microchannels-an experimental study, Int. J. Therm. Sci. 219 (2026) 110232. https://doi.org/10.1016/j.ijthermalsci.2025.110232.

[19] Q. Zhang, Z. Li, B. Li, J. Xu, H. Wang, Characterization of bubble population morphology in direct contact heat transfer process under different flow regimes and its influence on heat transfer performance, Exp. Therm. Fluid Sci. 147 (2023) 110933. https://doi.org/10.1016/j.expthermflusci.2023.110933.

[20] V. Garbin, D. Bothe, G. Brenn, C.M. Casciola, C. Colin, M. Marengo, F. Risso, G. Tryggvason, D. Lohse, Bubbles and bubbly flows, Int. J. Multiphase Flow 190 (2025) 105240. https://doi.org/10.1016/j.ijmultiphaseflow.2025.105240.

[21] A. Kitagawa, R. Kobayashi, P. Denissenko, Y. Murai, Natural convection heat transfer enhancement using bubble injection between vertical parallel plates, Int. J. Heat Mass Transfer 202 (2023) 123658. https://doi.org/10.1016/j.ijheatmasstransfer.2022.123658.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[22] B. Kichatov, A. Korshunov, K. Son, E. Son, Combustion of emulsion-based foam, Combust. Flame 172 (2016) 162-172. https://doi.org/10.1016/j.combustflame.2016.07.017.

[23] B. Kichatov, A. Korshunov, A. Kiverin, A. Saveliev, The role of explosive boiling in the process of foamed emulsion combustion, Int. J. Heat Mass Transfer 119 (2018) 199-207. https://doi.org/10.1016/j.ijheatmasstransfer.2017.11.116.

[24] B. Kichatov, A. Korshunov, A. Kiverin, E. Son, Foamed emulsion - Fuel on the base of water-saturated oils, Fuel 203 (2017) 261-268. https://doi.org/10.1016/j.fuel.2017.04.133.

[25] B. Kichatov, A. Korshunov, V. Gubernov, A. Kiverin, I. Yakovenko, Combustion of heptane-in-water emulsion foamed with hydrogen-oxygen mixture, Fuel Process. Technol. 198 (2020) 106230. https://doi.org/10.1016/j.fuproc.2019.106230.

[26] B. Kichatov, A. Korshunov, A. Kiverin, E. Son, Combustion of foamed emulsions in the quenching/reignition regime, Energy Fuels 31(7) (2017) 7572-7581. https://doi.org/10.1021/acs.energyfuels.7b00816.

[27] B. Kichatov, A. Korshunov, A. Kiverin, N. Medvetskaya, Combustion of foamed emulsion prepared via bubbling of oxygen-nitrogen gaseous mixture through the oil-in-water emulsion, Fuel Process. Technol. 186 (2019) 25-34. https://doi.org/10.1016/j.fuproc.2018.12.019.

[28] B. Kichatov, A. Korshunov, V. Sudakov, A. Golubkov, A. Kolobov, L. Chikishev, Magnetic nanofluid Fe_3O_4 in passive cooling system based on thermo-osmotic effect, Int. J. Therm. Sci. 199 (2024) 108931. https://doi.org/10.1016/j.ijthermalsci.2024.108931.

[29] B. Kichatov, A. Korshunov, V. Sudakov, V. Gubernov, A. Golubkov, A. Kiverin, A. Nastulyavichus, S. Kudryashov, Pattern formation and collective effects during the process of the motion of magnetic nanomotors in narrow channels, Phys. Chem. Chem. Phys. 25 (2023) 11780-11788. https://doi.org/10.1039/D3CP00435J.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[30] E.Y.M. Ang, P.C. Wang, W. Toh, T.Y. Ng, Effects of velocity on evaporation rate of nanoscale water droplets, Int. Commun. Heat Mass Transfer 164(Part A) (2025) 108851. https://doi.org/10.1016/j.icheatmasstransfer.2025.108851.

[31] G.R. Chandel, V.S. Sivasankar, S. Das, Evaporation of active drops: Puncturing drops and particle deposits of ring galaxy patterns, Phys. Rev. Fluids 9 (2024) 033603. https://doi.org/10.1103/PhysRevFluids.9.033603.

[32] B. Kichatov, A. Korshunov, V. Sudakov, Mechanism chaotic movement of Leidenfrost droplets, Exp. Therm. Fluid Sci. 159 (2024) 111277. https://doi.org/10.1016/j.expthermflusci.2024.111277.

[33] A.N. Yurasov, M.M. Yashin, D.A. Saifulina, T.N. Bakhvalova, Features of magnetooptical spectra of the transverse Kerr effect in CoPt nanostructures, Phys. Wave Phen. 32 (2024) 160-163. https://doi.org/10.3103/S1541308X24700109.

[34] S. Ouyang, Z. Wang, J. Wang, Q. Dong, J. Wang, Sessile droplet evaporation in the uniform electric field: A lattice Boltzmann method study, Phys. Fluids 37 (2025) 013101. https://doi.org/10.1063/5.0247410.

[35] S. Das, Low Reynolds number flows of active liquids in permeable, long, and thin capillaries, Phys. Fluids 36(9) (2024) 091702. https://doi.org/10.1063/5.0226605.

[36] B. Kichatov, A. Korshunov, V. Sudakov, D. Smovzh, O. Nerushev, Controlling the movement of active droplets in emulsion using chemically reactive magnetic microparticles, Phys. Fluids 37 (2025) 082144. https://doi.org/10.1063/5.0289062.

[37] P. Singh, P. A. Korevaar, Predator-prey behavior of droplets propelling through self-generated channels in crystalline surfactant layers, Angew. Chem. 137(24) (2025) e202502352. https://doi.org/10.1002/ange.202502352.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[38] Q. Zhang, J. Wu, J. Yan, S. Li, C. Li, Z. Han, D. Li, Y. Zhang, X. Wang, Q. Zhang, P. Zhang, Precise and large-scale droplet manipulation base on the stable interface triboelectric transfer, Surf. Interfaces 69 (2025) 106765. https://doi.org/10.1016/j.surfin.2025.106765.

[39] B. Kichatov, A. Korshunov, V. Sudakov, L. Chikishev, Cluster formation in active emulsion, Phys. Fluids 37 (2025) 042028. https://doi.org/10.1063/5.0265312.

[40] B. Kichatov, A. Korshunov, V. Sudakov, Movement of active droplets in dense emulsion, Phys. Fluids 36 (2024) 121709. https://doi.org/10.1063/5.0247172.

[41] J. Luo, M. Cui, X. Lian, B. Yuan, C. Song, Y. Ma, Y. Wang, One-step multiple emulsions driven by interfacial neutralization reaction, Langmuir 41(16) (2025) 10197-10204. https://doi.org/10.1021/acs.langmuir.4c05100.

[42] A. Yu, Z. Guo, Covalently grafted slippery surfaces for transport of large-volume droplets and bubbles, Small 21(25) (2025) 2412695. https://doi.org/10.1002/smll.202412695.

[43] B. Kichatov, A. Korshunov, V. Sudakov, V. Gubernov, A. Golubkov, A. Kiverin, Self-Organization of active droplets into vortex-like structures, Langmuir 37 (2021) 9892-9900. https://doi.org/10.1021/acs.langmuir.1c01615.

[44] B. Kichatov, A. Korshunov, V. Sudakov, V. Gubernov, A. Golubkov, A. Kiverin, Superfast active droplets as micromotors for locomotion of passive droplets and intensification of mixing, ACS Appl. Mater. Interfaces 13 (2021) 38877-38885. https://doi.org/10.1021/acsami.1c09912.

[45] S. Yuan, B. Yao, G. Yang, H. Lei, Light-driven multi-core/shell microdroplets as a targeted drug delivery system, Adv. Funct. Mater. 34(36) (2024) 2401905. https://doi.org/10.1002/adfm.202401905.

[46] B. Kichatov, A. Korshunov, V. Sudakov, V. Gubernov, I. Yakovenko, A. Kiverin, Crystallization of active emulsion, Langmuir 37 (2021) 5691-5698. https://doi.org/10.1021/acs.langmuir.1c00630.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[47] Y. He, C. Huang, J. Liu, X. Yang, F. Jiao, Capture of a moving ferrofluid droplet in a minichannel with a nonuniform magnetic field, Chem. Eng. Sci. 319 (2026) 122314. https://doi.org/10.1016/j.ces.2025.122314.

[48] S. K. Mani, S. Al-Tooqi, J. Song, A. Sapre, L. D. Zarzar, A. Sen, Dynamic oscillation and motion of oil-in-water emulsion droplets, Angew. Chem., Int. Ed. Engl. 63(6) (2024) e202316242. https://doi.org/10.1002/anie.202316242.

[49] Z. Zhao, J. Chen, G. Zhan, S. Gu, J. Cong, M. Liu, Y. Liu, Controlling the collective behaviors of ultrasound-driven nanomotors via frequency regulation, Micromachines 15(2) (2024) 262. https://doi.org/10.3390/mi15020262.

[50] R. Fuchs, S. Abdoli, M. Kilani, N.-A. Nor-Azman, R. Yu, S.-Y. Tang, M.D. Dickey, G. Mao, K. Kalantar-Zadeh, J. Tang, Stroking through electrolyte: Liquid metal droplet propulsion through pulse time modulation, Adv. Funct. Mater. 34(17) (2024) 2314815. https://doi.org/10.1002/adfm.202314815.

[51] B.V. Kichatov, V. Sudakov, A.M. Korshunov, D.V. Smovzh, M.S. Skirda, Micromotors Mn_3O_4@Fe_2O_3@CoO@ZnO@graphite for enhanced photocatalytic degradation of dyes, Colloids Surf., A 709 (2025) 136100. https://doi.org/10.1016/j.colsurfa.2025.136100.

[52] R. Huang, H. Luo, Z. Meng, W. Chen, R. Chen, Q. He, Y. Guo, L. Zhou, W. Wang, Q. Chen, Virus-like Cu_7S_4/vSiO_2/TA micromotors for localized bacterial capture and synergistic photothermal therapy, Colloids Surf., B 258 (2026) 115246. https://doi.org/10.1016/j.colsurfb.2025.115246.

[53] A.J. Martínez, M. Basharat, S. Chen, S. Sánchez, K Villa, Flow-active liquid marbles as microreactors for photocatalytic micromotors, Small 21(43) (2025) e05439. https://doi.org/10.1002/smll.202505439

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[54] J. Liu, R. Zhuang, D. Zhou, X. Chang, L. Li, Design and manufacturing of micro/nanorobots, Int. J. Extreme Manuf. 6(6) (2024) 062006. https://doi.org/10.1088/2631-7990/ad720f.

[55] T.-L. Xu, C.-R. Qin, B. Tang, J.-C. Gao, J. Zhou, K. Chen, T.H. Zhang, W.-D. Tian, Constrained motion of self-propelling eccentric disks linked by a spring, J. Chem. Phys. 161(6) (2024) 064905. https://doi.org/10.1063/5.0217158.

[56] B. Kichatov, A. Korshunov, V. Sudakov, A. Golubkov, D. Smovzh, S. Sakhapov, M. Skirda, The role of self-diffusiophoresis and reactive force during the propulsion of manganese-based catalytic micromotors, Phys. Chem. Chem. Phys. 26 (2024) 1612-1615. https://doi.org/10.1039/D3CP04689C.

[57] Y. Huang, Z. Lin, Q. He, Directional and reconfigurable assembly of active colloidal motors triggered by chemical communications, Adv. Funct. Mater. 34(6) (2024) 2311136. https://doi.org/10.1002/adfm.202311136.

[58] B. Kichatov, A. Korshunov, V. Sudakov, V. Gubernov, A. Kolobov, E. Korshunova, A. Kiverin, Oscillating motion of oil droplets in the emulsion near the air-water interface, J. Phys. Chem. B 125(36) (2021) 10373-10382. https://doi.org/10.1021/acs.jpcb.1c06128.

[59] B.V. Derjaguin, N.V. Churaev, V.M. Muller, Surface Forces (Springer Science+Business, New York, 1987).

[60] B. Kichatov, A. Korshunov, V. Sudakov, A. Golubkov, D. Smovzh, S. Sakhapov, M. Skirda, Magnetic manganese-based micromotors $MnFe_2O_4$@Fe_3O_4/graphite and Mn_2O_3@Fe_2O_3@Fe_3O_4/graphite for organic pollutant degradation, Adv. Sustainable Syst. 7(12) (2023) 2300288. https://doi.org/10.1002/adsu.202300288.

[61] J. Frueh, S. Rutkowski, T. Si, Y.- X. Ren, M. Gai, S.I. Tverdokhlebov, G. Qiu, J. Schmitt, Q. He, J. Wang, Direct measurement of thermophoretic and photophoretic force acting on hot micromotors with optical tweezers, Appl. Surf. Sci. 549 (2021) 149319.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

https://doi.org/10.1016/j.apsusc.2021.149319.

[62] S.S. Dukhin, B.V. Derjaguin, Electrophoresis, (in Russian) Nauka, Moskva (1976).

[63] B. Kichatov, A. Korshunov, V. Sudakov, A. Golubkov, V. Gubernov, A. Kiverin, Motion of a chemically reactive bimetal motor in a magnetic field, Phys. Chem. Chem. Phys. 24 (2022) 19693-19696. https://doi.org/10.1039/D2CP03383F.

[64] B. Kichatov, A. Korshunov, V. Sudakov, Chemical magnetism - surface force to move motors, Phys. Chem. Chem. Phys. 26 (2024) 24542-24552.
https://doi.org/10.1039/D4CP02537G.

[65] B. Kichatov, V. Sudakov, A. Korshunov, Movement of bimetallic motors under chemical magnetism force, Sens. Actuators, A 395 (2025) 117019.
https://doi.org/10.1016/j.sna.2025.117019.

[66] B. Kichatov, V. Sudakov, A. Korshunov, Motors based on photo-magnetic materials, J. Chem. Phys. 162 (2025) 244701. https://doi.org/10.1063/5.0271145.

[67] O.F. Petrov, R.E. Boltnev, M.M. Vasiliev, Experimental evolution of active Brownian grains driven by quantum effects in superfluid helium, Sci. Rep. 12 (2022) 6085.
https://doi.org/10.1038/s41598-022-09523-z.

[68] Y.M.F. El Hasadi, M. Crapper, Self-propelled nanofluids a path to a highly effective coolant, Appl. Therm. Eng. 127 (2017) 857-869.
https://doi.org/10.1016/j.applthermaleng.2017.08.050.

[69] Y.M.F. El Hasadi, M. Crapper, Self-propelled nanofluids a coolant inspired from nature with enhanced thermal transport properties, J. Mol. Liq. 313 (2020) 113548.
https://doi.org/10.1016/j.molliq.2020.113548.

[70] W. Peng, A. Chandra, P. Keblinski, J.L. Moran, Thermal transport dynamics in active heat transfer fluids (AHTF), J. Appl. Phys. 129(17) (2021) 174702.
https://doi.org/10.1063/5.0047283.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

[71] J. Velazquez, P. Keblinski, J.L. Moran, Active heat transfer fluids (AHTF): Enhancement of convective heat transfer by bubble-driven self-propelled microparticles, Int. J. Heat Mass Transfer 256 (2026) 127966. https://doi.org/10.1016/j.ijheatmasstransfer.2025.127966.

[72] B. Kichatov, V. Sudakov, A. Korshunov, Intensification of heat transfer using active bubble liquid, Phys. Fluids 37 (2025) 113304. https://doi.org/10.1063/5.0302299.

[73] K.J. Solis, J.E. Martin, Multiaxial fields drive the thermal conductivity switching of a magneto-responsive platelet suspension, Soft Matter 9 (38) (2013) 9182-9188. https://doi.org/10.1039/C3SM50820J.

[74] B. Kichatov, A. Korshunov, V. Sudakov, A. Golubkov, Evaporation of ferrofluid drop in magnetic field in Leidenfrost mode, J. Magn. Magn. Mater. 588 (2023) 171410. https://doi.org/10.1016/j.jmmm.2023.171410.

[75] P.J. de Visser, D. Karagrigoriou, A.-D.C. Nguindjel, P.A. Korevaar, Quorum sensing in emulsion droplet swarms driven by a surfactant competition system, Adv. Sci. 11(30) (2024) 2307919. https://doi.org/10.1002/advs.202307919.

[76] H. Gu, M. Li, J. Zhang, Z. Wang, Recent advances in strategies for inhibiting Leidenfrost effect, J. Therm. Sci. Technol. (Tokyo, Jpn.) 20(1) (2025) 24-00360. https://doi.org/10.1299/jtst.24-00360.

[77] L.-S. Chen, P.-I. Kung, Y.-S. Lin, Y.-X. Zhang, C.-C. Hsu, Influence of magnetic particle fluid flow field control on droplet evaporation, Appl. Therm. Eng. 279(Part E) (2025) 127866. https://doi.org/10.1016/j.applthermaleng.2025.127866.

[78] J. Li, P. Lv, Y. Cao, J. Ye, F. Li, C. Ma, L. Shi, N. Tan, Photothermal evaporation of the ferromagnetic nanofluid droplets under a magnetic field, Case Stud. Therm. Eng. 56 (2024) 104300. https://doi.org/10.1016/j.csite.2024.104300.

[79] E.M. Starinskaya, A.A. Rodionov, A.I. Safonov, V.S. Sulyaeva, N.B. Miskiv, S.V. Starinskiy, Formation of uniform spatial distribution of Fe_3O_4 nanoparticles during evaporation

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771

of a magnetic nanofluid droplet on a biphilic surface, Int. J. Heat Mass Transfer 239 (2025) 126602. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126602.

[80] S. Das, Thermally fully developed pipe flows of active liquids, Phys. Fluids 37 (2025) 032110. https://doi.org/10.1063/5.0258996.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5914771